

Harry Nguyen

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EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • GPA: 3.75/4.0

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

EXPERIENCE

TiMoto AI | Software Engineer

Sep 2025 – Present

- Designed and maintained **gRPC inter-service layer** with **exactly-once delivery** at **sub-50ms p99**; debugged production deadlock by tracing shared resource acquisition conflicts and redesigning call sequencing – zero data loss at production traffic
- Architected multi-AZ ECS Fargate with **Terraform IaC**, **circuit breaker pattern** with fail-open degraded mode, and CloudWatch + Prometheus + Grafana observability – **99.9% uptime**, **44% cost reduction** (\$40–60/month); auto-rollback on health check failure
- Led backend development for a 3-person engineering team – built **Python (FastAPI/Django) REST APIs**, **PostgreSQL + Redis** data layer, and **LLM API integration** (Claude, GPT) for production AI workflows at scale
- Architected **vLLM inference engine** with **PagedAttention** and **LLM-as-a-judge evaluation pipeline** – zero OOM failures at production traffic; automated quality measurement at **sub-50ms p99**

Google | Software Engineer Intern – Chrome Browser

May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with Protocol Buffers – shipped to Chrome stable channel serving **3B+ active users** at **sub-50ms p99**, **10K+ req/sec**; zero post-ship regressions
- Identified settings navigation as p99 bottleneck at 1,200ms; designed and implemented **lock-free concurrent trie search** eliminating mutex contention – **96% latency reduction**, zero production regressions
- Architected event-driven TypeScript/React system with observer pattern – **68% feature delivery acceleration** across **25K+ lines** of Chromium at **95% test coverage**; shipped to **3B+ active users**
- Delivered design documents and code reviews at **95% test coverage**; changes adopted into production branch reviewed by senior Chrome engineers

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on AWS – selected JWT over session auth for horizontal scalability; auto-scaling policy supporting **500+ concurrent users**, **90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned with **PostgreSQL indexing** achieving **sub-100ms** for 10,000+ records

PROJECTS

Pulumi – Contributor | Go, TypeScript, IaC

github.com/pulumi

- Submitted **Go CLI features and bug fixes** enabling multi-cloud (AWS/Azure/GCP) provisioning – under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer – studied linearizability guarantees and correctness under concurrent operations and partial failures

SKILLS

Distributed Systems: gRPC, Protocol Buffers, Exactly-once semantics, Circuit breakers, Fault tolerance, Kafka, Multi-AZ, Event-driven architecture

Languages: Python, TypeScript, Go, C++, Java, JavaScript, SQL, Bash, Rust

Frameworks & Databases: FastAPI, Django, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB

Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, Kubernetes, Docker, CI/CD, CloudWatch, Prometheus, Grafana

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LLM APIs (Claude, GPT), LangChain, LLM-as-a-judge evaluation, PyTorch

AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor